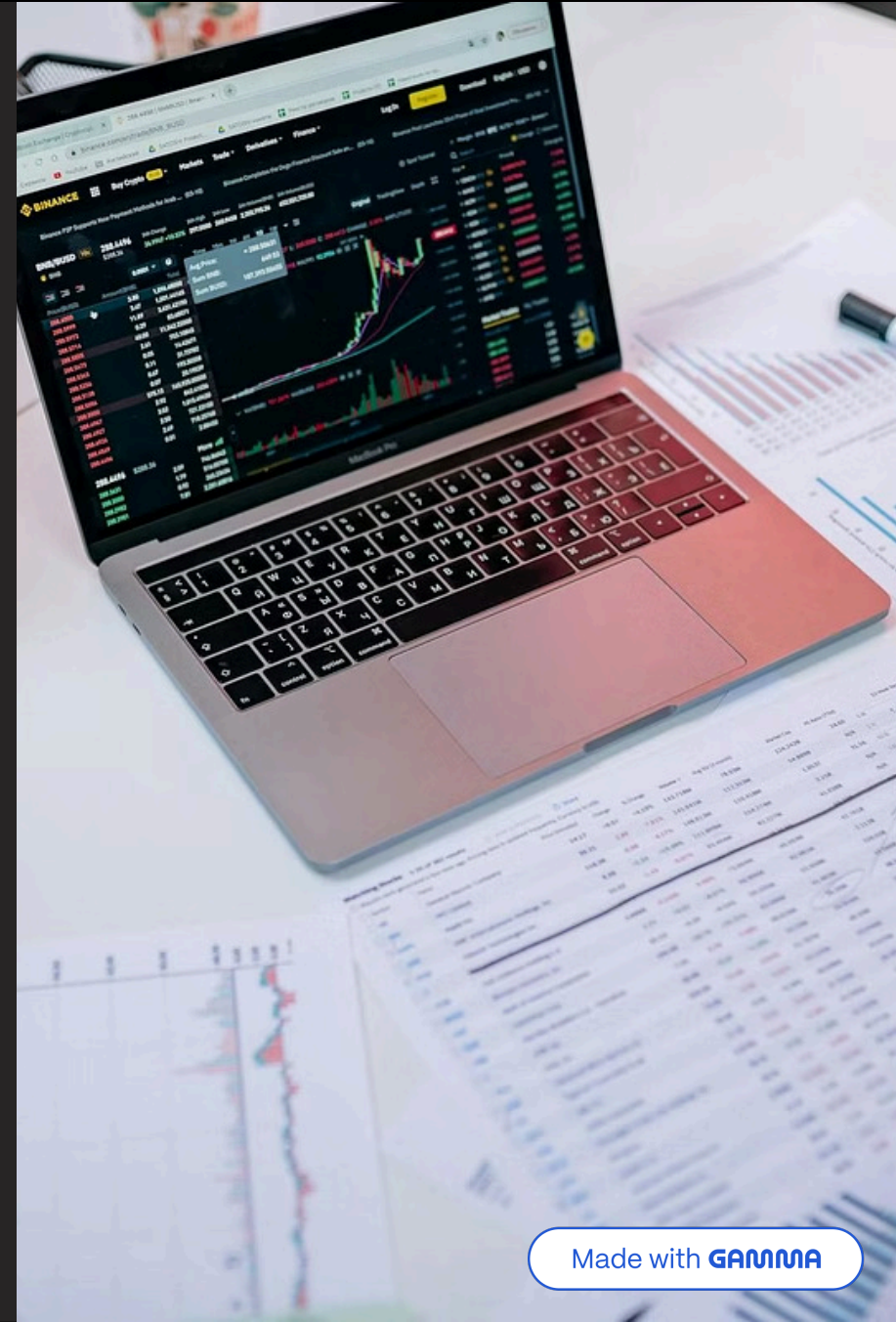


FinanceAI

Financial Portfolio Management MVP

Team Members: Riya Sharma (CS-2341760), Abhinav Singh (CS-2341754), Ayush Pandey (CS-23411152)

Faculty Advisor: Mr. Dibyanarayan Hazra



Problem Statement

Today's investors face significant challenges in managing their financial portfolios effectively. Without a centralized platform, they struggle to track investments, understand risk exposure, and make data-driven decisions.



Fragmented Tracking

Investments scattered across multiple platforms without unified visibility



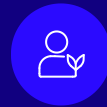
Risk Blindness

Limited awareness of portfolio risk exposure and volatility



Complex Analysis

Manual financial analysis consumes excessive time and requires expertise



Beginner Unfriendly

Lack of accessible portfolio insights for new investors

Project Objective

Our goal was to create a practical platform that combines portfolio tracking, intelligent analysis, and AI-powered recommendations into a single, accessible interface.

01

Secure Platform

Build robust portfolio management system with industry-standard security

02

Transaction Analytics

Enable CSV-based transaction upload with automated data processing

03

Risk Insights

Provide comprehensive portfolio risk analysis and exposure metrics

04

AI Recommendations

Offer intelligent stock recommendations based on portfolio analysis

05

Report Generation

Generate and export detailed portfolio reports for record-keeping

Technology Stack

Frontend

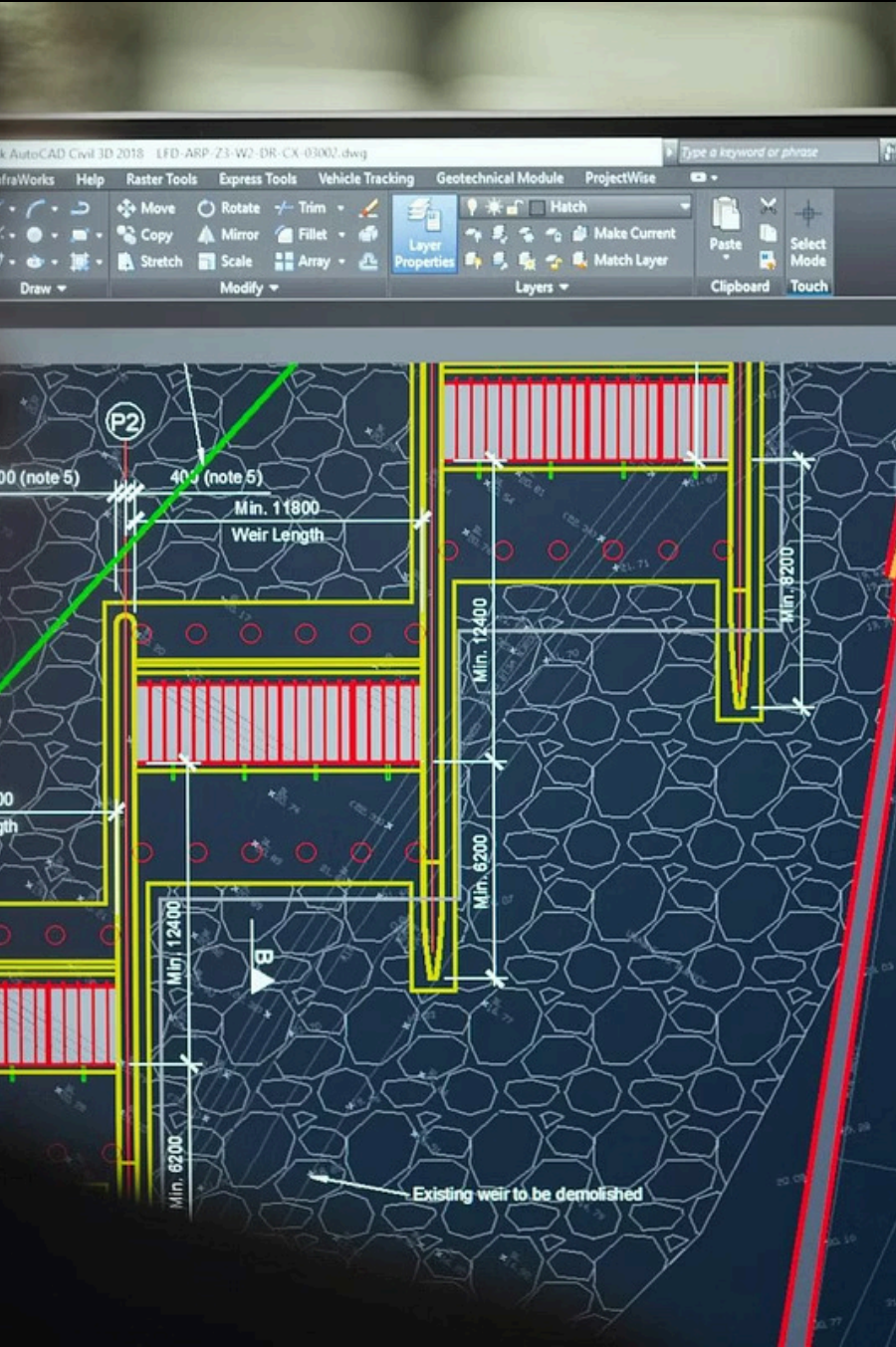
- React for component-based UI
- React Router for navigation
- Recharts for data visualization
- Modern CSS architecture

Backend

- FastAPI framework
- RESTful API endpoints
- Business logic implementation
- Request validation

Database & Security

- SQLite for data persistence
- Token-based authentication
- Password hashing
- Session management



System Architecture

User Interface Layer

React frontend for UI and state

Clear System Architecture

Data Layer

SQLite for reliable storage

API Layer

FastAPI handling business logic

Core Features



Secure Authentication

JWT-based token authentication with hashed passwords ensures account security



Portfolio Dashboard

Unified view of all investments, performance metrics, and key insights



CSV Import

Bulk transaction upload from CSV files for quick portfolio setup



Risk Analysis

Comprehensive risk assessment including volatility, diversification, and exposure



Stock Recommendations

AI-style suggestions for portfolio optimization and diversification



Report Generation

Export portfolio data and analytics reports via email or download

Database Design

The database schema is designed to efficiently manage user data, portfolio transactions, and analytical reports through normalized table structure.

users

User accounts and profile information

sessions

Active session tracking and management

transactions

Portfolio transaction history and details

watchlist

Tracked stocks and securities

uploads

CSV upload metadata and processing

reports

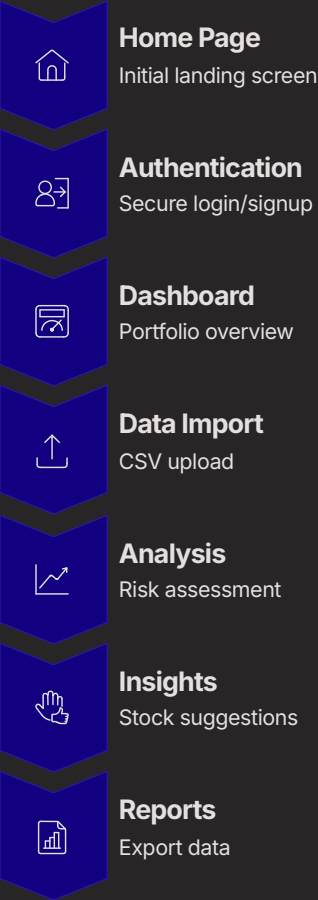
Generated portfolio reports

report_emails

Email notification tracking

Demo Flow

Our demonstration follows the complete user journey from account creation to portfolio insights.



Why This Is an MVP



Core Features Implemented

The platform delivers essential portfolio management capabilities including secure authentication, transaction tracking, risk analysis, and AI-style recommendations.

Production-Ready Foundation

Real user authentication, persistent data storage, and functional analytics flow demonstrate working software ready for real-world use.

Scalable Architecture

The modular design provides clear pathways for future enhancements while solving immediate portfolio management needs.

Future Scope & Conclusion

Future Enhancements

- **Live Market Data**
Real-time stock APIs for current pricing
- **ML Predictions**
Machine learning for price forecasting
- **Real-Time Valuation**
Live portfolio value updates
- **PDF Export**
Professional report formatting
- **Cloud Deployment**
Scalable infrastructure
- **Broker Integration**
Direct trading platform connections

Project Highlights

Secure

Token-based auth with hashed passwords

Practical

Solves real portfolio management problems

Scalable

Modular architecture for growth

Relevant

Real-world financial application